



Operation Manual

BTN3-5-8-12-15-20-30G/GW



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1. TERMS

Please read the manual before installation, use and maintenance, follow the instruction

These below three signals indicate important message.

	Warning ! It will be dangerous if you did not follow the instruction.
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	Notice! It will cause the damage of spare parts, burner even surrounding
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Notice!	Special message
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It is better that there has manual near the burner for necessary use

BTN3G/GW...30G Burner is natural gas with full-automatic sliding or modulation. It is a spare part for most heating equipment, such as water oven, steam boiler, air heater, etc.

- Fuel natural gas - heating value: $H_i = 35.8 \text{ MJ/m}^3 = 8550 \text{ kcal/m}^3$
- LPG heating value $H_i = 92.1 \text{ MJ/m}^3 = 22000 \text{ kcal/m}^3$
- Fuel is natural gas, or LPG.
- When the fuel is not natural gas, you have to learn its composition, if you are not sure about it, ask the supplier.
- If necessary, decrease gas pressure through installing pressure regulator. Gas consumption is up to gas pressure regulator and gas conjoined valve.
- There has motor in burner, supplying enough air. Make sure burner works fine by enough high and stable pressure.
- Sliding gas burner Max. controlling range is (GC series burner) 30%–100%
- Needed gas consumption: each 10KWh needs 13 m^3 air
- Controller controls burner operation automatically, detect the flame.
- Boiler temperature regulator and pressure regulator controls burner load.
- IP40
- control voltage 230V (-15%...+10%) 50Hz 1-phase
- Supply voltage: BTN400...1500GS/M: 380–400V 50Hz 3-phase
- The prerequisite of burner operation ambient temperature is 0. +40°C when working.

Check below items before first start burner:

- Connection (motor turning direction)
- Adjustment and controlling system set
- Boiler and other equipments under working.
- Burner can get enough air
- Open gas supply pipeline valve
- There has enough air in pipeline
- Gas pipeline leakage check is finished

Gas pressure is enough

	Notice! Burner must be installed fastenly, it causes the damage of burner and spare parts if shake.
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	Notice ! Exhausting in pipeline before first start
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	Warning! Gas leaks: - No ignition, no touch circuit equipments. - Close external main fuel stop valve. - Make sure that no person in leaking area. - There is ventilation in leaking area. - Take some measurements. On fire or other dangerous situations. - Cut circuit - Close external main fuel stop valve. - Take some measurements.
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	Warning! Do not use flame to check burner or boiler. It is forbidden to put flammable things in boiler room.
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	Warning! Close the boiler door when burner starts or works.
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It is best guarantee that burner has regular adjustment and correct installation.

Notice!	Follow local custom and rules when install and maintain gas burner or gas pipeline.
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It should be requested that the motor axis in level for burner installation, not allow to install the burner in opposite direction .

when you need some more spare parts for burner , please advise these below data.

· Burner model number

· System Code.

3. TECHNICAL DATA

3. 1. Briefness

Burner: (1) (2) (3) (4) (5)
BT N 3...30 G W

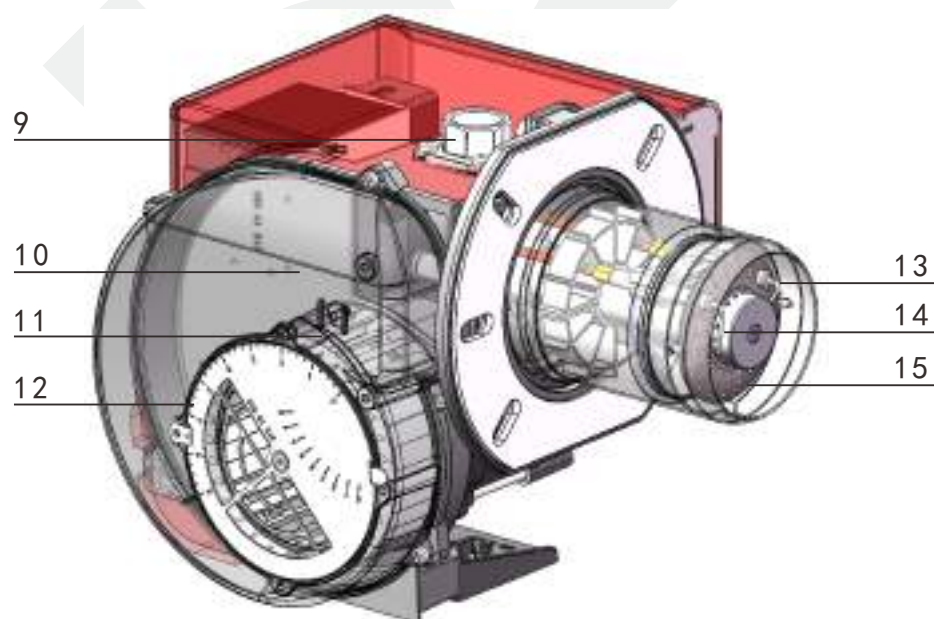
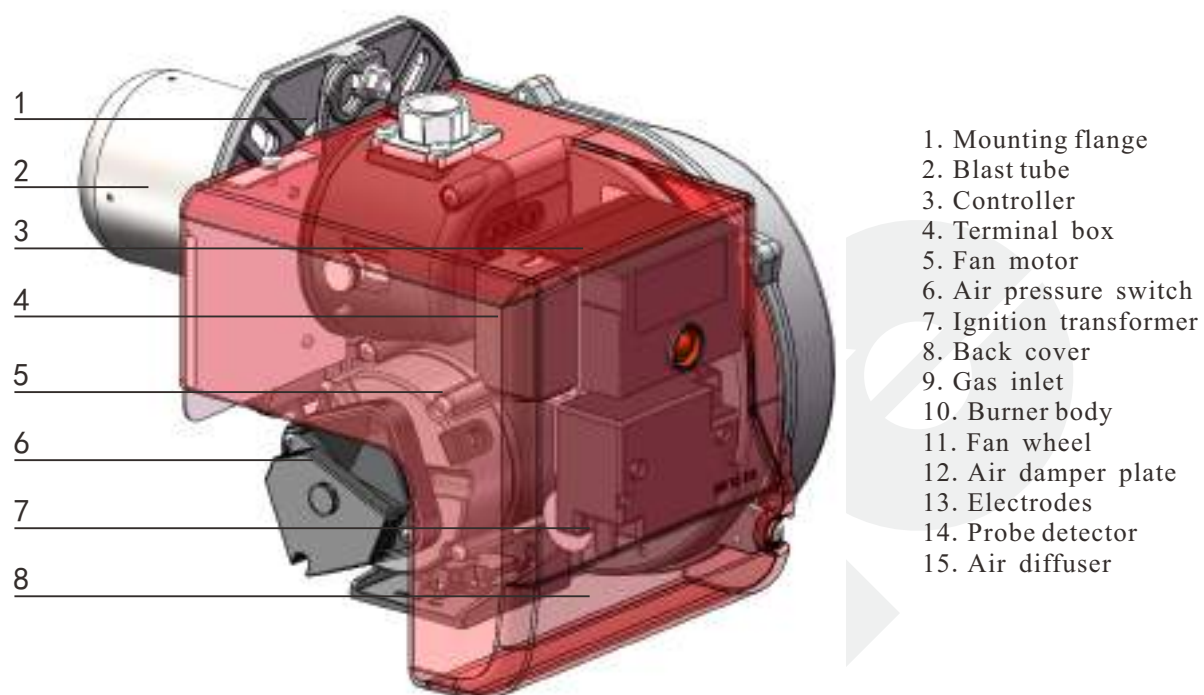
- (1) Company code
- (2) Series
- (3) Model No.
- (4) Gas
- (5) With silencer

- One stage fire
- Remove burner without removing flange.
- Use one flange and heat-proof gasket to connect with boiler
- Single phase fan motor
- Probe detector detects flame
- IP40
- Plastic protective cover

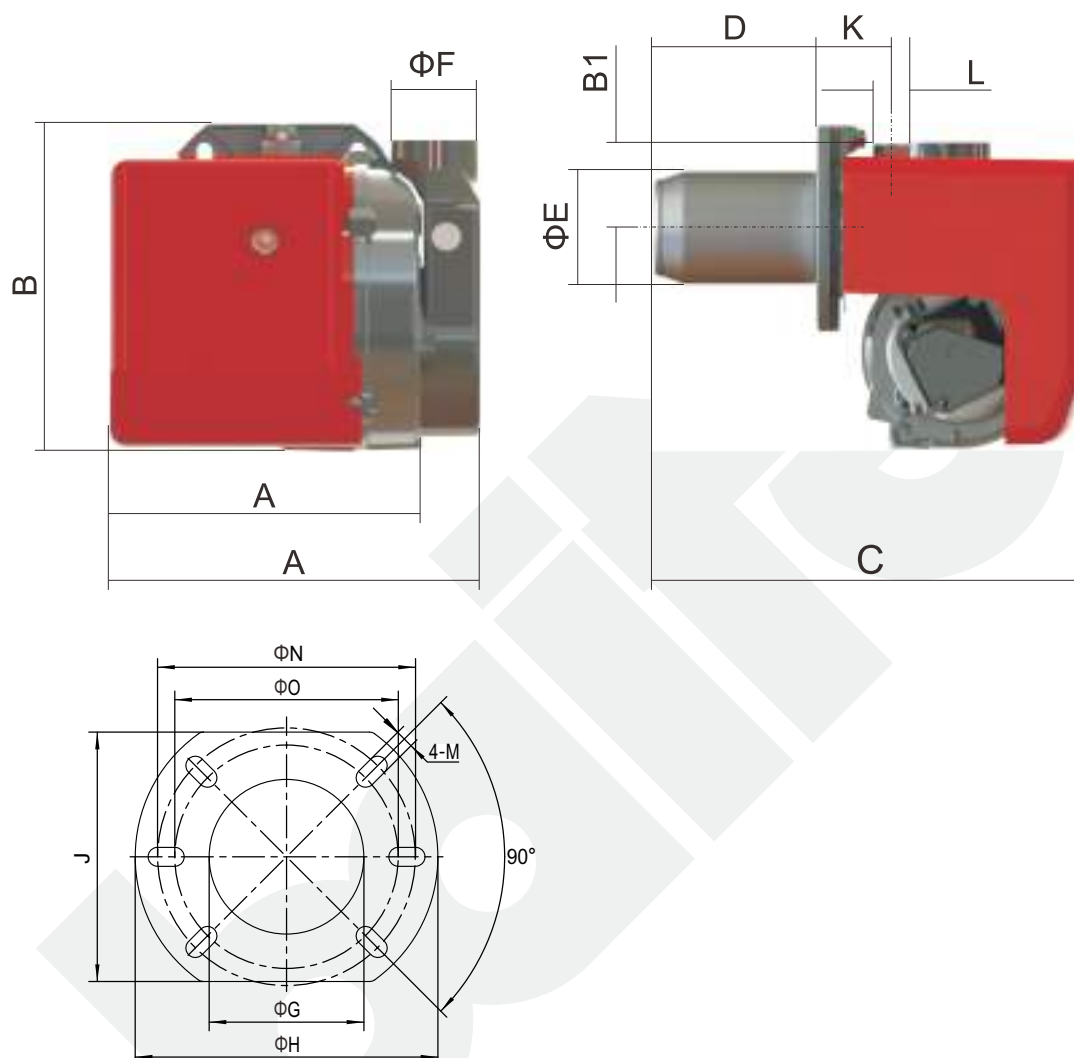
3. 2. Technical parameters

Burner	One stage fire						
	BTN3G/GW	BTN5G/GW	BTN8G/GW	BTN12G/GW	BTN15G/GW	BTN20G/GW	BTN30G/GW
capacity KW	17-48	34-70	49-108	49-120	50-160	60-205	100-280
Nm ³ /h	1.7-4.9	3.4-7	5-11	5-12	5-16	6-20.7	10-28.3
burner motor							
Voltage	230V	230V	230V	230V	230V	230V	230V
Frequency	50HZ	50HZ	50HZ	50HZ	50HZ	50HZ	50HZ
Output KW	0.09	0.09	0.1	0.13	0.2	0.2	0.25
CurrentA							
rpm	2850	2850	2850	2850	2850	2850	2850
Controller	LME	LME	LME	LME	LME	LME	LME
Flame detector	probe detector	probe detector	probe detector	probe detector	probe detector	probe detector	probe detector
Air damper	Manual	Manual	Manual	Manual	Manual	Manual	Manual
Weight	9	10	10	10.5	12	12	12.5

3.3. BTN3-5-8-12-15-20-30G/GW Standard Fittings Diagram



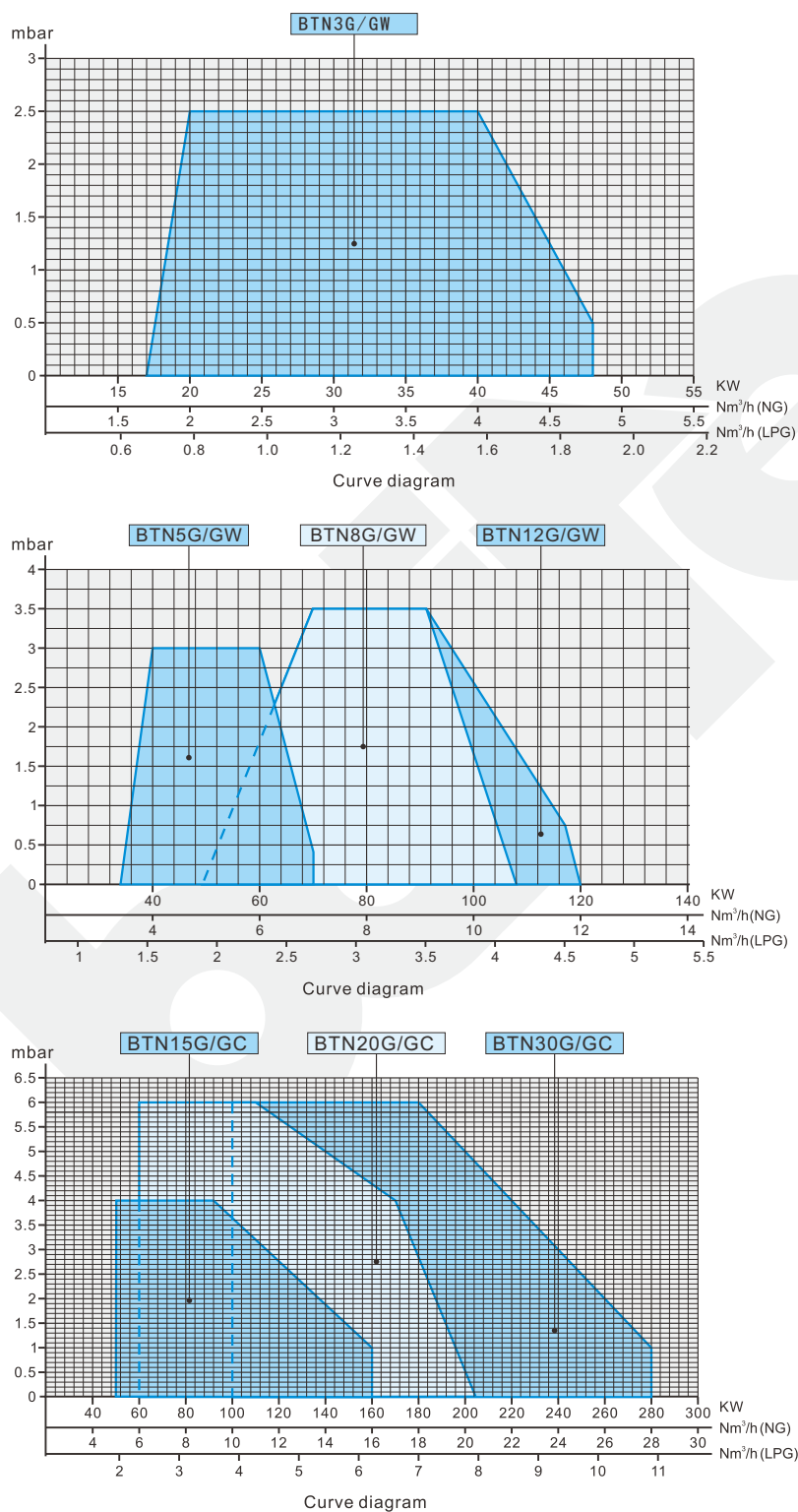
3.4. BTN3-5-8-12-15-20-30G/GW



Model	A	B	C	D	E	F	G	H	J	B1	K	L	O	N	M
	mm	mm	mm	mm	φ mm	φ mm	φ mm	φ mm	mm	mm	φ mm	φ mm	φ mm	φ mm	M
BTN3G	223	229	275	85	80	—	90	176	145	59.5	64.5	G1/2"	130	150	4-M8
BTN3GW	262	229	275	85	80	60	90	176	145	59.5	64.5	G1/2"	130	150	4-M8
BTN5G/GW	243/310	246	337	113	90	80	105	194	166	70	63	G3/4"	140	168	4-M8
BTN8G/GW	243/310	246	338	114	97	80	107	194	166	70	63	G3/4"	140	168	4-M8
BTN12G/GW	243/310	246	338	114	97	80	107	194	166	70	63	G3/4"	140	168	4-M8
BTN15G/GW	290/357	302	433	160	114	—	124	220	195	70	72	G1"	160	190	4-M8
BTN20G/GW	290/357	302	433	160	125	—	135	220	195	70	72	G1"	160	190	4-M8
BTN30G/GW	290/357	302	433	160	125	—	135	220	195	70	72	G1"	160	190	4-M8

Note: "G" is furnace wall hole dimension, Mounting hole centre dimension refer to O or N.

3.5. Woring Range Curve Diagram



Comparing with burning output of combustion chamber pressure that is the Max. value under ideal test. All data is base on 20°C air temperature and 500meters altitude .

Blast tube length is customized

4.BURNER INSTALLATION

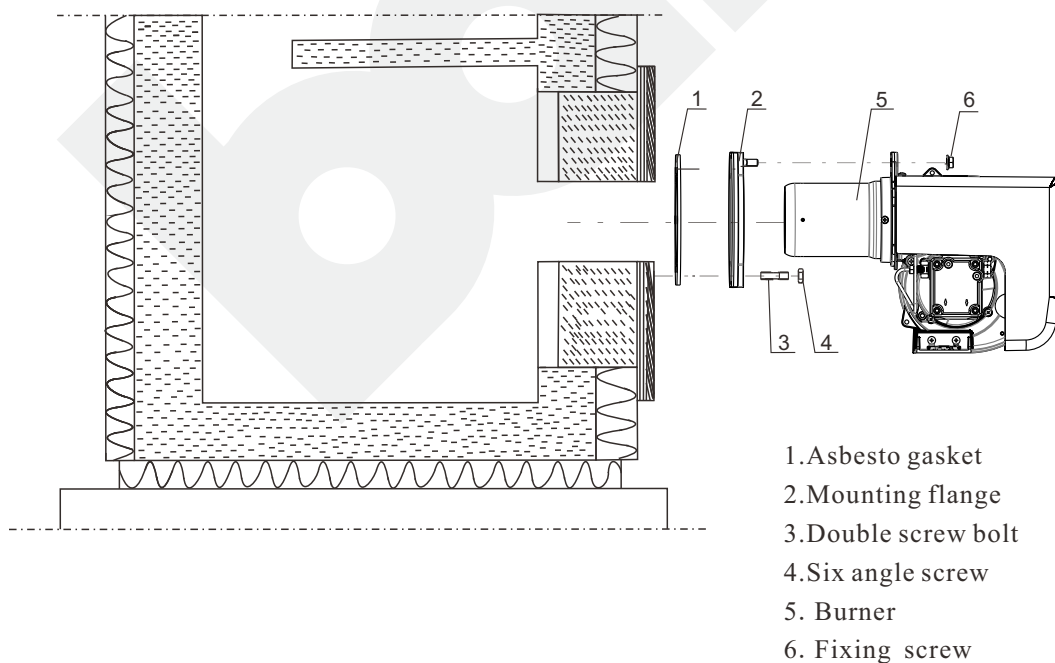
4.1. BURNER INSTALLATION

Preparation before installation

1. Check chimney (sectional area and height) .
2. Voltage and frequency.
3. Gas system and dimension ,gas pipeline valve voltage stabilizer and accessory seal.
4. Check burner accessory.
5. Reduced pressure voltage stabilizer is behind filter
6. Rust clear inside pipe

Installation

1. Put gasket inside boiler between mounting board and flange,use double head bolt and six angle screw to fasten,After taking burner head and tube into combustion chamber,fasten the bolt , fix burner. There must be seal among boiler board,gasket and mounting flange.no air leak in case that the heating smoke will reduces heating reaction or damage spare parts during running.
2. Connect gas valve to burner properly.
3. Connect circuit.



Note: Installation board must have standard burner screw thread hole please install burner refer to diagram.

4.2.CIRCUIT CONNECTION

Burner connection must follow what supplier` wiring diagram told.

4.3. SUPPLY GAS PIPELINE INSTALLATION

The dimension of supply gas pipe which locates behind the pressure regulator must bigger or equal to valve.

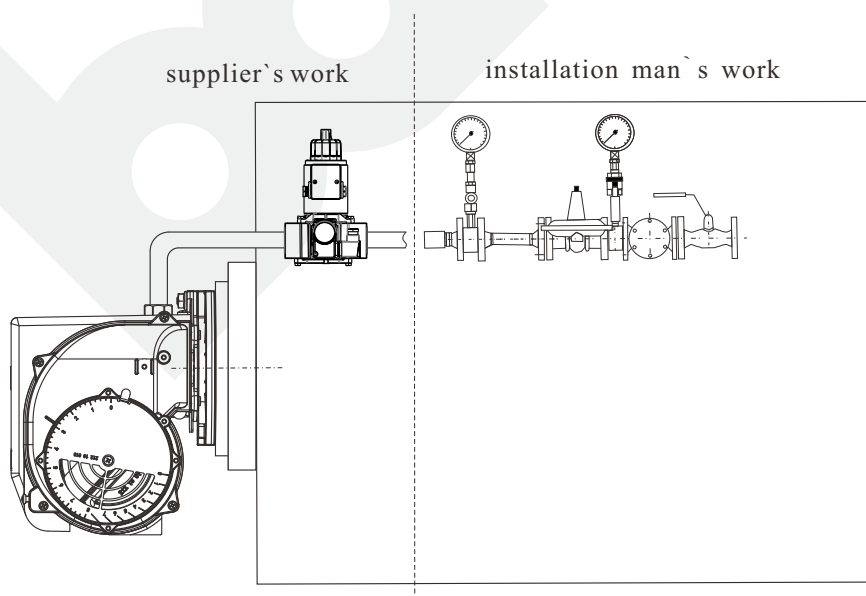
Note! There must have manual stop valve before burner adjustment equipments.



Notice! There must have alone filter before gas valve

Gas pipe ventilation

Connecting flexible hose, open gas valve pressure inlet for exhaust. when ventilates, open gas pipeline ball valve, full pipeline gas , then close the inlet.

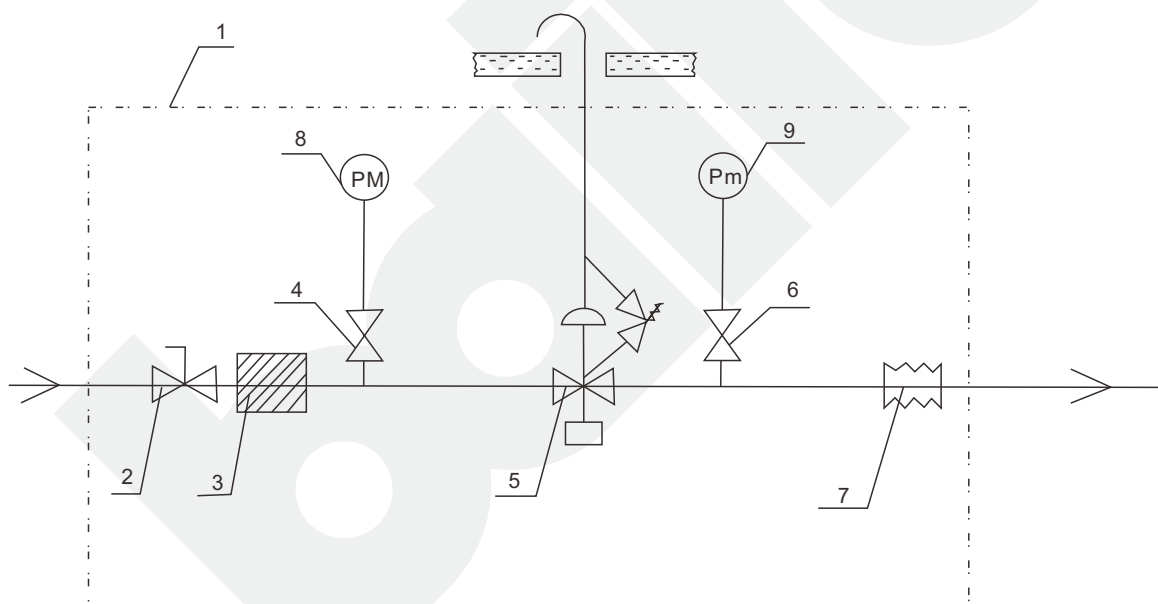


4. 4. GAS PRESSURE ADJUSTMENT EQUIPMENTS

If input gas pressure is higher the original set P_{max} , it must be reduced . If the pressure is not stable, adjust it . If there has no safety blowoff valve and safety stop valve in pressure regulator, install them .

Safety blowoff valve must have correct installation to guarantee that safety valve will not release under the condition that burner will close as full load with main gas line interrupt.

Safety stop valve should be set to close when gas pressure about 60% more than second pressure (the pressure is adjusted by regulator) . but not higher than P_{MAX} . Safety should be set to open as pressure is 30% higher than second pressure. The selection of pressure regulator is affected by inlet pressure, second pressure, gas flow and its kind .



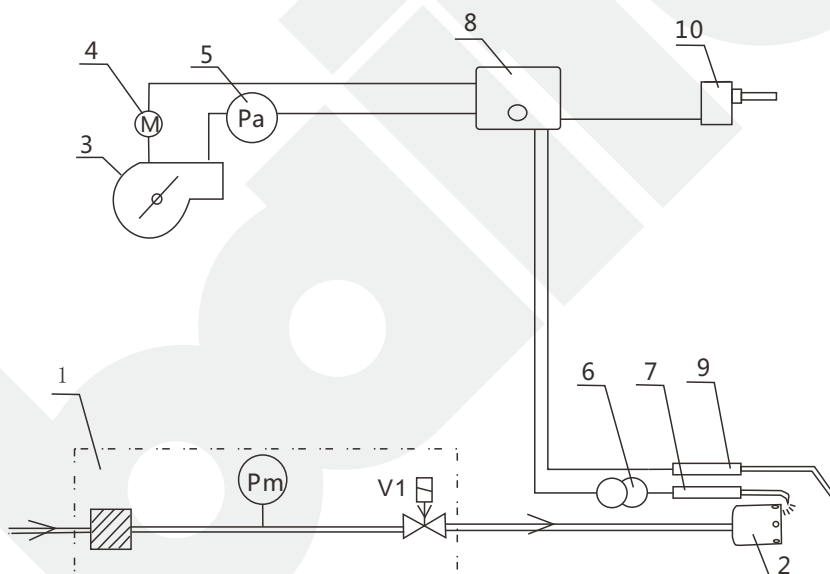
- | | |
|---|--|
| 1. Gas pressure regulator | 6. Gauge valve |
| 2. Shutoff valve | 7. Bellows compensator/flexible connection |
| 3. Gas filter | 8. Gauge, high pressure |
| 4. Gauge valve | 9. Gauge, low pressure |
| 5. Regulator with safety stop valve and blowoff valve | |

5.BURNER OPERATION PRINCIPLE

5.1.Operation Process of BTN3-5-8-12-15-20-30G/GW

Burner starts, controller(8) release start signal. motor(4) runs (time delays/LME11)。
 prepurge begins. air pressure switch(5) watches motor pressure. When prepurge finishes, transformer working arc forms in electrode(7), gas valve(V1) opens, gas is sent to gas combined nozzle(2), then ignited, probe detector(9) begins to feedback signal to controller until ends. burner is under controlling equipment device(10) watch.

N o t e !	Burner will not start and stop work if pressure does not exceed pressure switch setting value.
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- | | |
|-----------------------------------|--|
| 1. Gas valve | 5. PA Air pressure (differential pressure switch) |
| - Gas filter | 6. Transformer |
| - Pressure switch, low limit min. | 7. Electrode |
| - Gas valve V1, NC | 8. Controller |
| - Flow regulator | 9. Probe detector |
| 2. Gas combined nozzle | 10. Controlling equipment (temperature regulator/pressure regulator) |
| 3. Fan motor | Pm. Gas pressure (low pressure) |
| 4. Burner motor | |

5.2. START AND RUNNING

Note Burner motor is single phase 220V .Check Plus-n-Minus connection for ignition. Exhausting in the gas pipe before burner connects to it. Make sure that keep windows and doors open.Opening pipeline elbow which locates near burner .then open a little of gas stop valve ,close the valve when smell the gas .When exhaust finishes,burner connects gas pipeline again.then do following measurements.

1. Check carefully of burning result emission(Boiler gas brake and chimney is open) and the water in boiler
2. Adjusting flame combustion-supporting air..The burner air valve indicates servo motor.the air gap between burner head and diffuser will be open 1/3
3. Properly adjusting gas valve flow adjustment equipment,until to needed flow.
4. Connecting main power.then controller will works under process.After switch on ,during prepurge, transformer will connect power if air presssure switch detects enough pressure which exceeds setting value,consequently ,gas valve is connected (safety valve and running valve) each valve is open fully,gas output is limited by the setting position which is manually adjusted by inner flow adjustment in running valve(main valve).First ignition will come out “dead fire”result.

Here are the causes:

- a) Gas pipes do not exhaust enough air. then gas flow is too less to keep flame stable.
- b). “Dead fire” maybe the flame is not stable in ionization zone or UV photocell does not detect flame, air/gas proportion not correct.then change air or gas flow to correct. Rotate burner head adjustment equipments, close the air pipe between burner head and gas pipe, or open a little.
- c). Ionization current possibly is effected by transformer released discharging current blocks.
(two kind of currents will go to same way through burner ground line) , then burner will stop thanks to little ionization. Change the transformer two wires connection to solve this problem.
Above problems sometimes are caused by bad connection of ground line.
- d) When burner works,watch gas flow meter(read it per minute),along with output flow adjustment to natural gas $=8550\text{kcal/m}^3$).this value by 60 is flow output per hour.This output flow can be revised by inner flow regulator of valve,adjust gas and relative air flow consumption.Throught professional tool to check burning condition.Gas and air flow ratio should be (Co2) in natural gas at least8% at lowest flame ,highest10%. CO in smoke is not over0.1%(1000 p.p.m)highest limit After adjustment,must close and reopen burner.

- e). Then, burner offers highest supply according to boiler request. Using professional tools to check burning situation. If necessary, change (air and gas adjustment ($\text{CO}: 95\text{mg}/\text{m}^3$ $\text{Nox } 200\text{mg}/\text{m}^3$)).
- f) Air pressure switch is that when air pressure is not correct, it will block gas valve open, so pressure switch must adjust to make the burner air pressure enough to close circuit. Pressure switch connection has self-control function which close as fan motor is off (there has no air pressure in burner). If air pressure switch does not detect the over limit pressure (no adjustment), controller does its recycle, but transformer does not connect the power and ignition valve is not open, the controller leave "stop". To guarantee pressure switch working, must raise adjustment value when burner is on small fire. when it comes to intervention points, burner goes to stop. Press button to release it, and adjust the pressure switch to detect enough air in pre-air process.
- g). Control gas pressure switch (highest_or_lowest) is to stop burner operation when gas pressure is not in the range. from the special function of pressure switch, the lowest pressure control switch uses contact when the pressure value exceeds preset one; the highest one uses it when lower, It is forbidden that controller connects power while adjusting gas pressure switch, Burner is on working, any gas pressure switch interruption lead to burner stop immediately. Check pressure switch when burner first starts.
- h). BY removed electrodes wires, burner starts, check photocell (probe detector). Controller must finish self recycle, and goes to "stop" status after 3s flame forms, this test does not be neglected when burner works. Remove wire of probe detector, controller must "stop" right away. If there has UV photocell, take it out after 1 mint fire. When UV photocell remove from its brace, can not "see" flame UV radiation. burner will come to "lock". If photocell is a bit dirty, it will effect its function. When photocell detecting head got dirty by oil, clean it completely, take care of finger movements also will influent the photocell performance. UV photocell can not "see" sun light or normal light. Check photocell sensitivity via lighter, candle or spark of transformer electrodes. To make excellect work, UV photocell current should be stable and can not reduce to the value which controller requests. maybe need to make photocell move in board finding the best position.
- i). Check boiler thermostat or pressure switch function (must stop burner firstly)

6. BURNER ADJUSTMENT

6.1 Load regulator

6.1.1 BRAHMA or DUNGS gas valve

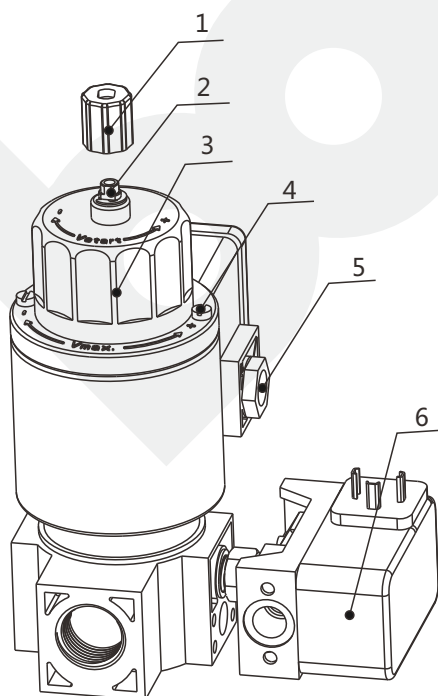
Model:EG....or MVDLE.....(one stage fire burner)

Gas pressure switch -

Gas pressure switch is set 10Mbar.If necessary,remove transparent cover,set adjustment fan wheel.The error of scale plate setting value and real one is $\pm 15\%$. then recover the cap

Load regulator

Loose cylinder screw(4),check adjustment.If necessary,regulating gas flow by rotating hydraulic flow regulator(3).Rotate clockwise(-)=reduce gas flow.rotate anticlockwise(+) =increase gas flow, refasten screw (4) Fast open stroke adjustment Remove seal cap(1),use back side as tool. Rotate fast/slow open speed adjustment rod (2) to requested direction until get needed gas flow.Along (-)direction,slowly open speed,(+)direction, fasten open speed(gas valve),recover the seal cap(1).Setting value ,start route is adjusted from close position 1/4-1/2 cycle.



characteristic :

- Valve normally in close status
- With flow regulation
- Slow open fast close
- It is adjustable of valve opens

- 1.Seal cap
- 2.Fast/slow open speed regulation rod
- 3.Flow regulator
- 4.Cap screw (no paint)
- 5.Power terminal
- 6.Gas pressure switch

N o t e !	Do not open paint seal locking screw
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6. 1. 2. DUNGS GAS VALVE

Model : MB-DLE...(one stage fire burner)

Air pressure switch

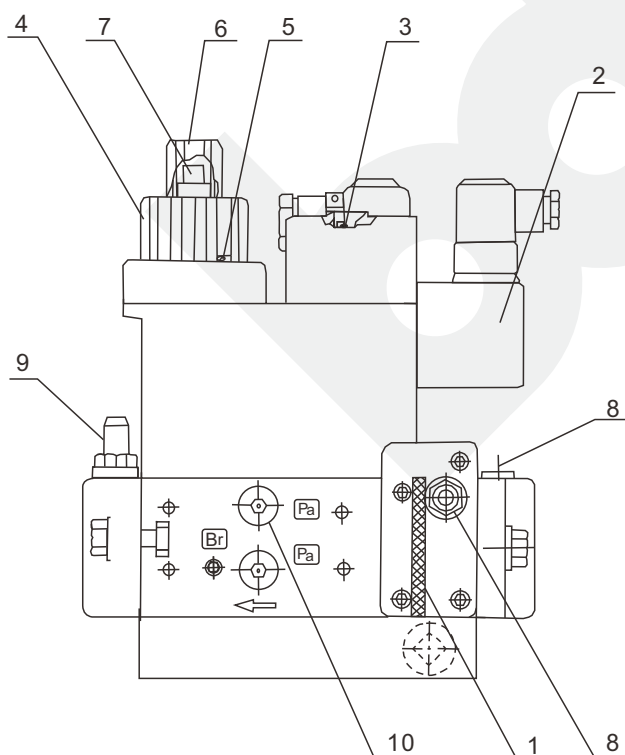
Air pressure switch is set by supplier 10mbar. If necessary, remove transparent cover, set adjustment fan wheel switch point. The allowable error between scales plate set value and real is $\pm 15\%$.

Load regulator

Loose cap screw(5) one cycle, check regulation. If necessary, Regulate gas flow by rotating flow regulator(4). Rotate clockwise (-)=Reduce gas flow, Anticlockwise (+)= increase gas flow, Refasten cap screw(5). In order to regulate output flow, it needs to rotate pressure setting screw(3) with wrench regulating output pressure(=nozzle pressure). Output pressure range 4-50m bar, rotate clockwise, increase output pressure, anticlockwise (-), reduce it. output pressure(=nozzle pressure) is set by supplier, 30 cycles from close position.

Fast open stroke regulation

Remove seal cover(1), use back side as tool, rotate fast/slow open speed regulation rod(7) to requested direction until reach needed airflow. Rotate along direction(-), open speed slowly, along(+) rotate, quicken open speed(gas valve), reinstall the seal cap(1). Start process is regulated from close position 1/4-1/2 cycle.



characteristic :

- Valve normally is in close status.
- Regulator with flow, stabilizing pressure
- Slow speed open double fast close(safety valve)
- Valve open route can do the regulation

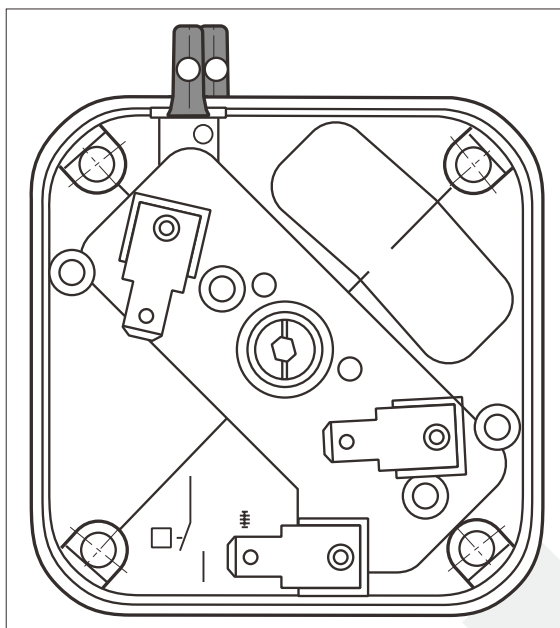
- 1.Filter
- 2.Gas pressure switch
- 3.Pressure setting screw
- 4.Flow regulator
- 5.Cap screw (without paint)
- 6.Seal cover
- 7.Fast/slow open speed regulation rod
- 8.Gas input pressure testing inlet
- 9.Gas output pressure testing outlet (nozzle pressure)
- 10.Gas testing terminal between two valves

N o t e !

Do not open the paint seal to lock screw .

6.1.3. Pressure switch

1. Air pressure switch LGW3A1



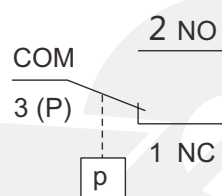
LGW...A1 Switch parameter

When pressure increase:

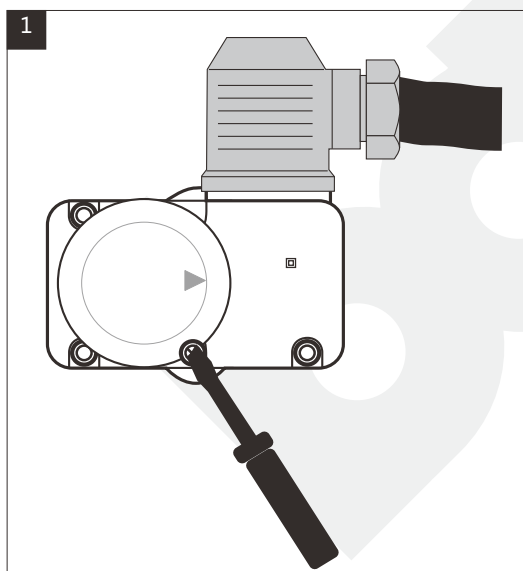
1 NC open, 2 NO close

When pressure decrease:

1 NC close, 2 NO open



2. Gas pressure switch GW...A...



Adjusting gas pressure switch

Use proper tool to remove cap cover,
screw driver 3 or PZ 2 fig 1
remove cap cover

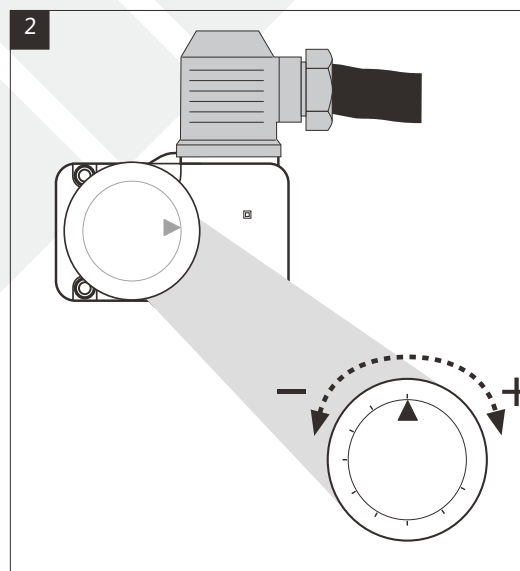
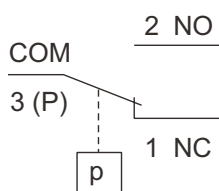
GW...A... Pressure parameter

When pressure increase:

1 NC, 2 NO

When pressure decrease:

1 NC 2 NO



Adjustable cam with scale to make the
pressure switch to set value.
fig 2

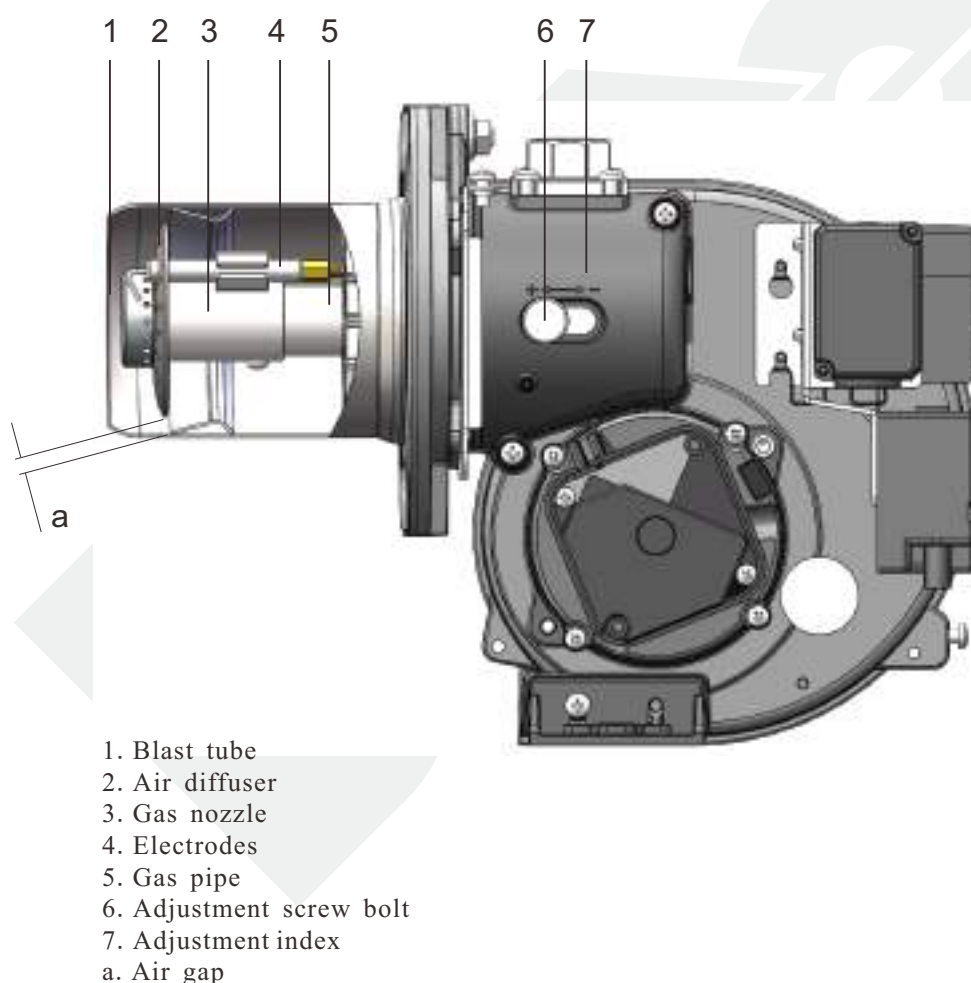
Pressure switch moves when pressure decreases
adjust to ▲
reinstall cap cover

6.2. BURNER HEAD ADJUSTMENT

Burner head air flow adjustment

By loosening adjustment screw bolt(6), make air diffuser(2) moves following adjustment index(7) direction, to change the distance between blast tube(1) and air diffuser(2) edge gap(a). In order to adjust air speed and consumption. If air diffuser(2) is so close to “-”, blast tube air speed will higher, it causes hard ignition or no enough air to support burning. If (2) close to “+”, then it will lower, it causes bad burning quality (high CO). So use smoke tester to check burning value.

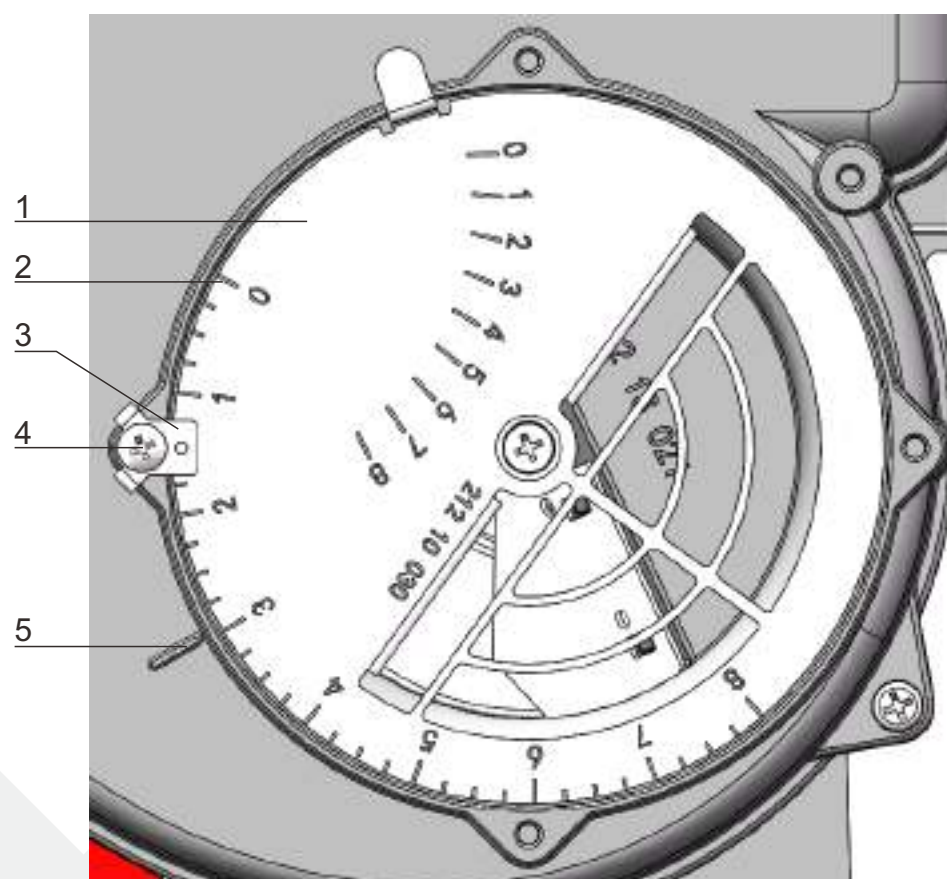
In real use, air diffuser(2) and blast tube(1) should be in middle position. .



NOTE!	If air diffuser position has changed, there changes the air speed in blast tube. Use smoke tester to check burning value. If necessary, have enough burning air consumption.
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6. 3. BURNING AIR ADJUSTMENT

6. 3. 1. Burner model number: BTN3-5-8-12-15-20-30G

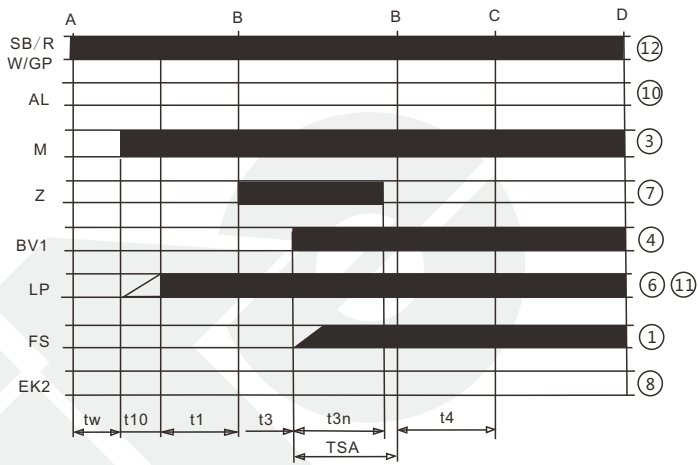
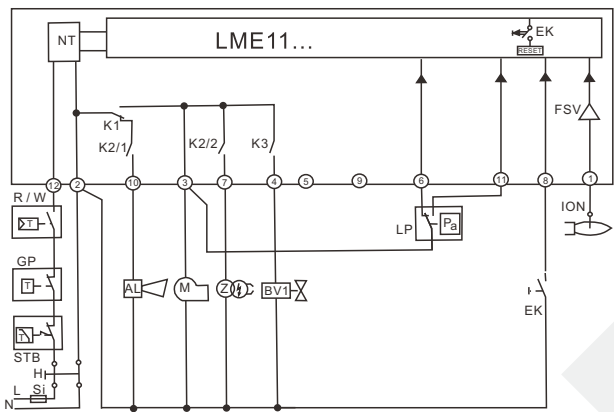


1. Air damper plate
2. Position scale value
3. Fixed press sheet
4. Fasten screw
5. Position index

Loose fasten screw (4), rotate air damper plate (1) position index (5), if the relative position scale value (2) is big, means that the air damper open large. vice verse. when correct adjustment, fasten the screw (4).

7. CONTROLLING SYSTEM

Controller LME11 function diagram(controlling system)



ON/OFF times

Flame detector model No.	controller	Voltage	tw approx. s	t1 min. s	TSA max. s	t3n approx. s	t3 approx. s	t4 approx. s	t22 approx. s ²⁾	t10 min s ³⁾	t11 min s ¹⁾	t12 min s ¹⁾
Probe detector (ION)	LME11.330A2	AC230V	2.5	30	3	2	2	—	—	5	—	—
Probe detector (ION) or photocell QRA... ⁴⁾ and auxiliary equipment ARQ3...	LME22.232A2	AC230V	2.5	20	3	2	3	8	—	3	16.5	16.5

Diagram

tw	Waiting time
TSA	Safety time
t1	Prepurge time
t3	pre -ignition time
t3n	Post ignition time
t4	《off》 and 《Bv2》 time interval
t10	Air pressure signal testing time
t11	Actuator 《SA》 sets start time
t12	Actuator 《SA》 sets close time
t22	Safety time

- 1) Actuator runs the longest time have to shorter
- 2) Flame controlling time 间
- 3) Max. 65s
- 4) Only use 230V AC

Function

- The prerequisite before start
- Burner controller resets
 - Close all contactors in line under thermal request.
 - Voltage without low voltage
 - Air pressure switch LP must locates in non load position or Cp1, or terminal 2
 - Fan motor or AGK25 is closed
 - Flame detection area is dark, no outer lightness

- Low voltage
- When rated voltage is low, it can cut safely from running position. AC175V (at UN=AC230V)
 - When rated voltage exceeds AC185V (at UN=AC230V), it needs to start again.

- Control batch-type operation
- Burner controller closes safely after below 24 hours ongoing operation, then start again.

- Reversed polarity protection
- If live line (terminal 12) and neutral line (terminal 2) is mixed, controller will start lock process after TSA finishes.

The controlling process in failure and lock indication

- If stop work, fan motor and ignition will be discharged (<1s)

Cause	Reaction
Main part fault	Restart
Voltage is lower than limit voltage	Close safely
Voltage is higher than low voltage limit	Restart
During «t1», there comes outer lightness	Lock
During «t1», there comes outer lighnesst	Avoid starting ,lock after 30s
«TSA» no find flame	LME11. . . at most 3 times repeat, then after «TSA» locks LME2... After «TSA»,lock immediately
Flame disappears in runing	LME11. . . • Build flame, «TSA» at most 3times repeat • No flame, TSA locks LME2... Lock
«Lp» closes in working position	Avoid starting, lock 65s
«Lp» close in normal position	«t10» stops when finishes
«t10» finishes, no air pressure signal.	Lock
During «tw», «CPI» is in open position	Avoid starting, lock after 60s

IF happens no work, LME still lock , red signal light is on . controller will lock immediately.
This condition also is used for part fault .

Reset controller

When stop works, controller will reset right away. press lock reset 1s (<3s) , LME will reset under
all pipeline contactor will be closed and no exits low voltage

Repeat times limit(only for LME11.)

If after «TSA», there has no flame, or flame being on , Control start passing «R» at most 3times
each time or other stop work will restart.

Running status indicator

During running, the happened status indicator refer to below table

The code table with color is suitable for kind of color signal light (LED)		
Status	Color Code	Color
Waiting time TW other waiting status		No
Ignition status, bad ignition		Yellow light flashes
Runing, ignition		Green light
Running, bad ignition		Green light flashes
There has outer lightness when burner runs.	▲ ▲ ▲ ▲ ▲	Green red light
Low voltage	▲ ▲ ▲ ▲ ▲	Yellow red light
Fault , alarm	▲	Red light
Fault code output	▲ ▲ ▲ ▲ ▲	Red light flashes
Interface Diagnosis	▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲	Light red light flashes

Diagram

...stable

No

▲ Red
Yellow
Green

Technical parameters

Rated voltage	AC120V=10%-15% AC230V=10%-15%
Rated frequency	50. . . 60Hz%
Electrical power consumption	12VA
External fuse (Si)	Max.10A(slow)
Installation position	Selectable
Terminal 12 input current	MAX.5A
Weight	about 160g
IP	I
IP	Ip40
Terminal 1 allowed cable length	Max.1m , 100pF/m (Max.3m , 15pF/m)
ORA...fitting AGQ3...A27 cable length (lay separate cable)	Max.20m in 100pF/m
Remote reset	Max.20m in 100pF/m
Terminal8 and 10 allowed cable length	Max.20m in 100pF/m
Other terminal allowed cable length	Max.3m in 100pF/m

Allowed ampere	At $\cos \phi \geq 0.6$	At $\cos \phi = 1$
-Terminal3	Max 2. 7A (15A in fastest0. 5s onlyLME2. . .)	Max 3A
-Terminal4, 5 and 7	Max.1. 7A	Max.2A
-Terminal10	Max 1A	Max. 1A

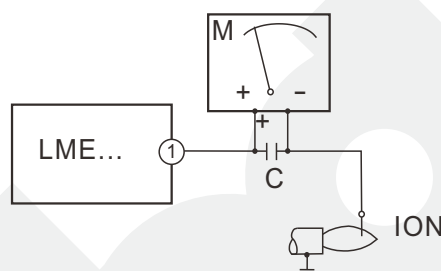
Using probe detector to check flame

	Rated voltage UN=AC ¹ 230V)
Detector voltage passes probe detect and ground, (AC voltage, Ri 10MΩ	AC 115...240V
Change start (setting value) : Open (flame is on) (DC , Ri ≤ 5kΩ) Close (flame is off) (DC , Ri ≤ 5kΩ)	≥ DC 1.5 μ A ≤ DC 0.5 μ A
The current of detector	≥ DC 3 μ A
Flame is weak when starts (green LED flashes)	Approx. DC 5 μ A
Short circuit of current between probe detector and ground (AC ammeter Ri 5kΩ)	Max. AC 100. . . 300 μ A

¹) In order to apply for use in outside Europe, rated voltage is AC 230V ± 10%

Notice : At same flame, LME...controller current maybe is lower than LMG..and LGB...
Flame detection is worked by transmission and revision . Flame signal amplifier only reacts for component part DC current of flame signal, probe detector and ground short circuit will lock the burner.

Measured current



Pic.

C Electrolysis capacitor 100. . . 470 μ A; DC 10...25V
ION probe detector

M Microammeter , Ri Max. 5000Ω

Check below items when burner is in maintenance

matters need attention

	Safety equipment	Anticipation reaction
a)	Probe detector line returns open before burner starts	LME11.... At most repeat 3times LME2.... 《TSA》 finishes , lock
b)	Burner flame is off simulately, then close gas supply	LME11.... 《TSA》 finishes flame at most 3times. TSA no finishes flame --lock LME2.... lock
c)	Simulate that air pressure fault when burner is running	Lock immediately



Warning! This controlling system is complete one! Not allow to change it.

8. MAINTENANCE



Warning! Cut off power before burner maintenance, also close gas supply pipe manual stop valve. When checking , close burner power, but open gas supply pipe manual stop valve.

BURNER MAINTENANCE

Make sure the good working , check electrodes position and clean.

Check flame detector position and status.

If necessary, clean the filter.

Check gas pipe

Clean dirty and float waste, keep burner clean

Do smoke test to check burning value regularly

HEATING EQUIPMENTS CONTROL

Keep boiler clean and door closed

Guarantee heating system has enough water pressure

Clean fire oven and chimney regularly

Check smoke board adjustment regularly

Avoid water spray into burner

Boiler must open vent fan

Check gas pipe and burner safety equipments regularly

Check boiler work

Notice: check electrodes

9. FAILURE AND MEASUREMENTS

Failure, firstly check below items :

1. Check circuit (control and supply voltage)
2. Check valve leakage detector (Yellow light is ok) ?
3. Check all adjustment device and controller installation.
4. The working of safe devices.
5. Check the burner if has fuel and fuel pipe valve is open or not ,gas pipe!

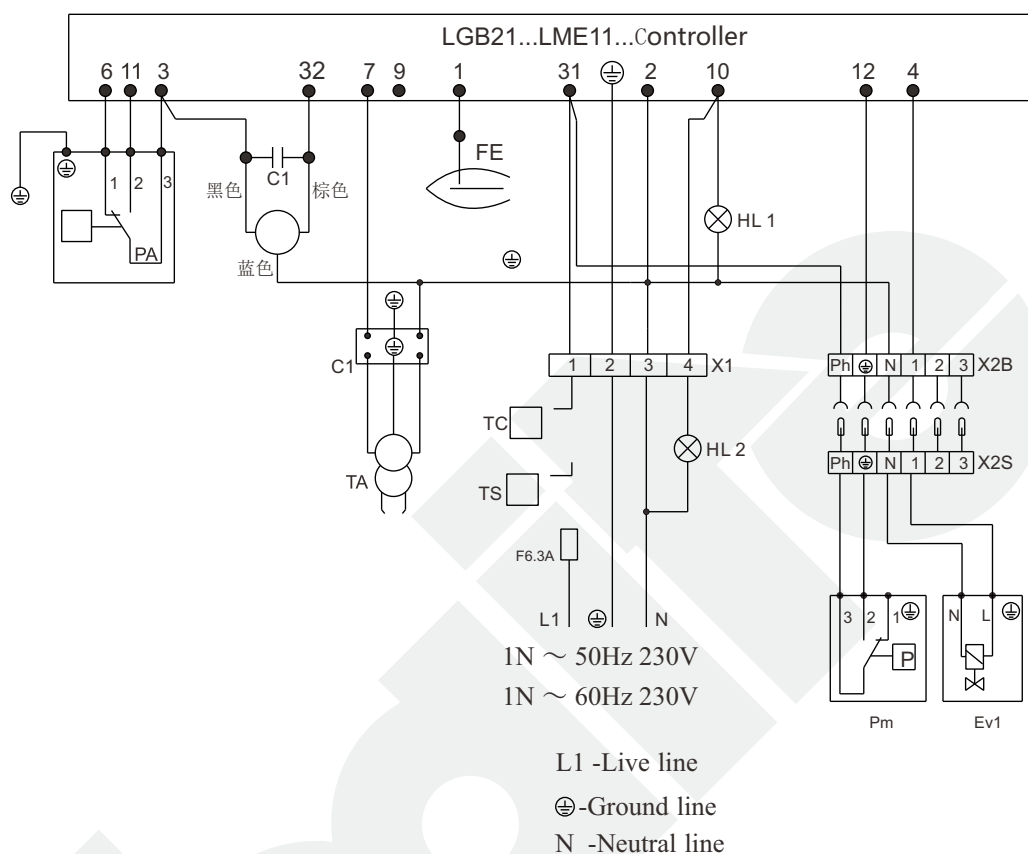
If failure isn't caused by above items, have to check

"Lock" (signal light is on) , need to reset.

Notice	Causes	Measurements
1. motor		
Motor can not start	Controlling system cut off -Controller failure	Find out the cause and revise it. Change controller
	Motor failure	Change
2. Air pressure is not enough		
Burner motor begins to work, but lock after finishing prepurge.	Air pressure switch installation is wrong.	Check installation , if necessary , change it.
	Hose is dirty	Clean
	Air pressure switch fault	Change
	Fan motor is dirty	Clean
3. Ignition failure		
Burner motor start, controlling voltage from controller to transformer is open no flame, lock.	Servo motor failure or wrong setting.	Change or adjustment
Burner motor start, controlling voltage from controller to transformer is closed, no flame, lock.	Electrodes are dirty or damaged, no insulation.	Clean or change
	Electrodes far from diffuser.	Adjustment
	Ignition cable damage	Change
	Ignition transfer failure.	Change

Notice	Causes	Measurements
4. No flame		
Burner motor statrm,spark comes out,after a short while, lock.	Gas valve is not open -Servo motor fails -Wire cable damages -Control return line is faulty -Air damper board servo motor cam switch set is wrong -Air damper servo motor fails Gas flow is wrong	Change damaged parts Adjustment Adjustment
5.Lock after flame forms		
Flamecomesoutthenstop running,airpressureswitch lowlimitrestart	Air pressure is too low -Pressure regulator fails Filter blocks Air pressure switch set is wrong(low limit)	Repair or change regulator Clean filter Adjustment
6. Flame detector failure(=lock)		
Prepurge lock	Flame detector failure	Change
	Controller failure	Change
Fanmotorstartscomesout theflame.lock	Flame detector position is wrong Flame detector is dirty	Adjustment Clean
Burner stop work and lock	Flame is weak Flame detector failure Controller failure	Check burner adjustment device Change Change
	Controller failure	Change

10. WIRING DIAGRAM OF BTN3-5-8-12-15-20-30G/GW



BTN3-5-8-12-15-20-30G/GW Wiring Diagram

Pm, Gas pressure switch

Pa, Air pressure switch

HL2, External fault indicator light

EV1, Gas solenoid valve

TA, Transformer

LME11, Controller

FE, Probe detector

C1, rejecto

6.3A, Fuse

TS, Safety thermostat

X1, Wire terminal board

M, Fan motor

Tc, boiler thermostat

HL1, Fault indicator light

X2B, X2S, 4pin socket

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